<!– marginalia.md – exercises Lout's @LeftNote / @RightNote / @OuterNote / @InnerNote machinery via raw-Lout fences.

The right margin is widened to 5.5cm (the default 2.5cm is too cramped for legible margin notes; the Lout User's Guide explicitly recommends this) and the page-headers line is suppressed so the eye does not have to compete with running heads.

Margin notes are NOT a markdown shorthand in mdlout; they live entirely in raw-Lout fences embedded mid-paragraph. The note attaches to the word immediately preceding the @RightNote / @LeftNote token. Lout then positions the note in the margin at the same vertical height as that word – shifting downward if a previous note would overlap, but never forward onto the next page. –>

Marginalia and Side-Notes

This page is set up with an extra-wide right margin (5.5 cm against the default 2.5 cm), which gives Lout's `@RightNote` machinery room to lay out readable annotations alongside the main column. The pattern is classic Tufte: keep the eye on a single column of prose, and let commentary, definitions, and references ride in the margin without breaking the flow.

The composite trapezoidal rule estimates the integral of a function by summing the areas of trapezoids fitted under the integrand on a uniform grid. It converges at second order in the step size for sufficiently smooth integrands.

Named for the polygon you get when you join successive samples of `f` *with straight lines.*

The note above appears flush against the right margin, set in italic at 80% of the body size (Lout's default `@MarginNoteFont`). The body line itself is not interrupted: Lout reflows the column as if the note were not there, and only the margin column knows it exists.

Multiple notes on one page

A second paragraph, with two notes:

Simpson's 1/3 rule goes one degree higher: it fits a quadratic to each triple of consecutive samples and integrates the quadratic exactly. The resulting estimator has *fourth-order* accuracy in the step size whenever the integrand is four times continuously differentiable on the interval.

Sometimes just called *Simpson's rule.* The 1/3 distinguishes it from the variant that uses a degree-three polynomial on four nodes – Simpson's 3/8 rule – which is mostly of historical interest.

Halving the step size cuts the error by a factor of sixteen, not four.

When two notes are close enough to overlap, Lout shifts the second one downward so they sit cleanly. You can see this in the rendered output: the second note above starts a line or two lower than the word it annotates.

Footnotes vs side-notes

Markdown's `[^name]` footnotes go to the foot of the page; side-notes go to the side.

Both have their place. Footnotes are the right choice for citations and substantive asides[2][1]; side-notes are the right choice for *micro-glosses* – a short definition, a reminder, or a running commentary the reader can ignore without losing the thread.

standard `[^name] -> @FootNote` pipeline; the side-notes elsewhere on the page go through raw-Lout `@RightNote` fences instead.

Layered annotation: an outer-margin gloss

`@OuterNote` is the binding-aware variant: it puts the note in the *outer* margin – the right margin on a recto (odd) page, the left margin on a verso (even) page. On a single-page test document like this one recto/verso don't differ, so `@OuterNote` is visually identical to `@RightNote`; the value of the macro shows in a multi-page bound book.

The continuous extension of the harmonic series to complex arguments is called the Riemann zeta function and its non-trivial zeros are the subject of one of the seven Millennium Prize Problems.

Riemann's 1859 memoir is eight pages long and contains the still-unproven conjecture about the location of its non-trivial zeros.

A note inside a numbered list

Side-notes work inside structured environments too. Here is a short numbered list, each item with its own marginal annotation:

1. Locate the integrand and the interval of integration.
2. Choose a step size `h` or, equivalently, a number of panels `n`.
3. Apply the rule and obtain the estimate.

On computer algebra systems the integrand often arrives already in symbolic form; the integration interval may not be obvious if the user has assumed implicit bounds.

Smaller `h` *means more function evaluations and lower truncation error.*

The estimate is exact in floating-point only up to round-off in the summation; for very large `n`*, use Kahan compensation.*

Closing observation

Margin notes are one of the features that justify having a real typesetting engine underneath mdlout: TeX, Lout, and InDesign all do this well, while every web-stack approximation involves CSS contortions and a non-flowing column. Lout's stream model handles the column reflow for free.

Tufte's *Visual Display of Quantitative Information* is the canonical reference for the side-note idiom. Knuth's *Concrete Mathematics* is a second touchstone: every page is structured around a wide right margin that carries running commentary, jokes, and reference pointers in a hand several decimal sizes smaller than the body. The same effect is possible here with `@RightNote`, a fixed `right-margin: 5.5c` in the frontmatter, and no further machinery.

[1]Like this one. Footnotes are rendered by mdlout's